	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19 (Autonomous Institute, Affiliated to VTU) **Computer Science and Engineering**

CIE- I		
CourseCode: 23CS4PCSED	**Course Title: Software Engineering**	
Semester: 4^{th} A,B,C,D,E,F,G	**MaximumMarks: 40**	**Date: 27/3/2025**
Faculty Handling the Course:	MRP, Dr.LNR, RM, SAK, MJ,RH, SVA	
Instructions: i) ***Answer all Part A and Part B Questions*** ***ii) In Part C - Answer 3a or 3b and 4a or 4b***		

PART –A (5 marks)

No.	Question	Marks	CO	PO	BL
1	As an expert in computer security, you have been approached by an organization that campaigns for the rights of torture victims and you have been asked to help them to gain unauthorized access to the computer systems of a company that are into selling equipment used directly in the torture of political prisoners. Discuss the ethical dilemmas that this request raises and the professional and ethical responsibilities of software engineer that you would be violating. State Code of Ethics and Professional Practice as specified by ACM/IEEE-CS joint task force.	5	1	8	3

PART –B (15 marks)

No.	Question	Marks	CO	PO	BL
2a	Consider an interactive system that allows users to check real-time traffic conditions from kiosks installed at various city locations, identify the Non-functional requirements required for such a system and explain using hierarchy diagram.	5	2	2	4
2b	Identify the various problems that could arise when requirements are written in natural language.	5	2	2	4
2c	Develop a Software Requirement Document for Library Management system as specified by IEEE/ANSI.	5	2	2	4

PART –C

No.	Question	Marks	CO	PO	BL
3a	i) Illustrate the different metrics used for specifying non-functional requirements. ii)Design a template using structured natural language to capture the requirements of a system that is used for dispensing cash in a bank ATM.	4+6	3	3	5
	OR				
3b	i)Describe evolutionary development model and explain why programs that are developed during evolutionary development are likely to be difficult to maintain. ii) Design a sequence diagram for a system showing interaction for article printing.	4+6	3	3	5
4a	i) Write the scenario for carrying out online payment using credit card for a purchase of a book. ii) Giving reasons for your answers based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: a) An interactive system that allows flight passengers to find flight times from terminals installed in airports. b) The spelling checks and correcting function in a word processor. c) A system to control anti-lock braking in a bike.	4+6	2	2	4
	OR				
4b	i)Suggest who might be the stakeholders in a university student record system. Justify, why it is almost inevitable that the requirements of different stakeholders will conflict in some way. ii) Identify and Analyze the principal viewpoints which might be taken into account and organize these using a view point hierarchy diagram for a online food ordering and delivery system	4+6	2	2	4

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Scheme and Solution - CIE-1

PART-A

1. As an expert in computer security, you have been approached by an organization that campaigns for the rights of torture victims and youhave been asked to help them to gain unauthorized access to thecomputer systems of a company that are into selling equipment used directly in the torture of political prisoners.
Discuss the ethical dilemmas that this request raises and the professional and ethical responsibilities of software engineer that you would be violating. State Code of Ethics and Professional Practice as specified by ACM/IEEE-CS joint task force.

Answer: professional and ethical responsibilities 2M, Code of Ethics -3M

The responsibilities that the software engineers should have towards professional and society

Confidentiality: Engineers should normally respect the confidentiality of their employers or clients irrespective of whether or not a formal confidentiality agreement has been signed.

Competence: Engineers should not misrepresent their level of competence. They should not knowingly accept work which is out with their competence.

Intellectual property rights: Engineers should be aware of local laws governing the use of intellectual property such as patents, copyright, etc. They should be careful to ensure that the intellectual property of employers and clients is protected.

Computer misuse: Software engineers should not use their technical skills to misuse other people's computers. Computer misuse ranges from relatively trivial (game playing on an employer's machine, say) to extremely serious (dissemination of viruses).

ACM/IEEE principles

PUBLIC: Software engineers shall act consistently with the public interest.

CLIENT AND EMPLOYER: Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.

PRODUCT: Software engineers shall ensure that their products and related modifications meet the highest professional standards possible

JUDGMENT: Software engineers shall maintain integrity and independence in their professional judgment.

MANAGEMENT: Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

PROFESSION: Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.

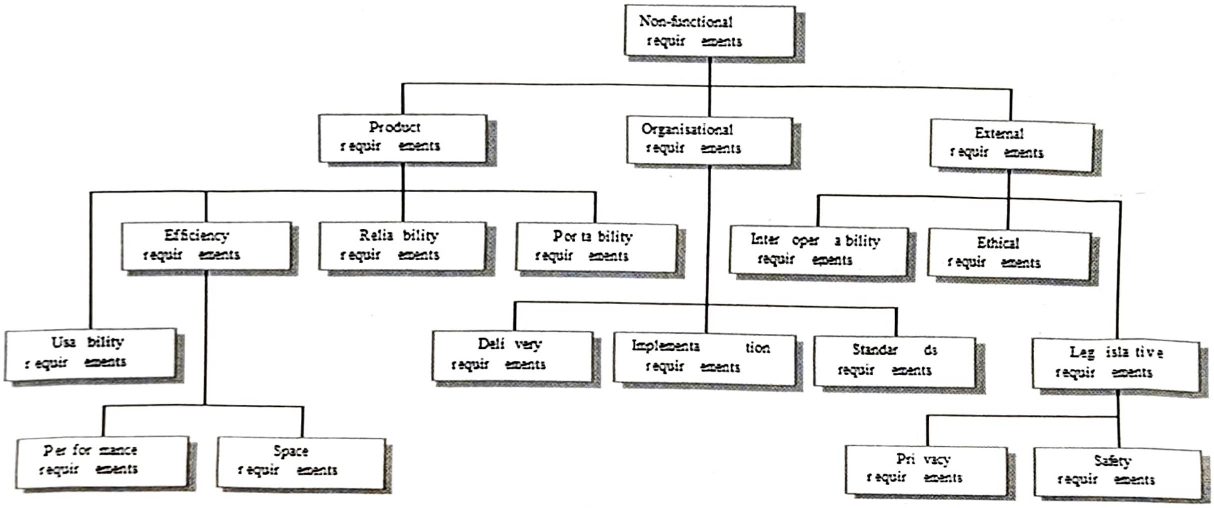
COLLEAGUES: Software engineers shall be fair to and supportive of their colleagues.

SELF: Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical

PART-B

2a. Consider an interactive system that allows users to check real-time traffic conditions from kiosks installed at various city locations, identify the Non-functional requirements required for such a system and explain using hierarchy diagram.

Answer: Explanation 2M, Hierarchy diagram -3M



2b. Identify the various problems that could arise when requirements are written in natural language.

Answer: Explanation 5M, (Any 5)

Problems of using natural language for defining user and system requirements

Ambiguity: The readers and writers of the requirement must interpret the same words in the same way. NL is naturally ambiguous so this is very difficult.

Over-flexibility: The same thing may be said in a number of different ways in the specification.

Lack of modularisation: NL structures are inadequate to structure system requirements.

Lack of clarity: Precision is difficult without making the document difficult to read.

Requirements confusion: Functional and non-functional requirements tend to be mixed-up.

Requirements amalgamation: Several different requirements may be expressed together.

2c. Develop a Software Requirement Document for Library Management system as specified by IEEE/ANSI.

Answer: Explanation 5M, (Any 5)

IEEE requirements for Library Management system

Defines a generic structure for a requirements document that must be instantiated for each specific system.

- Introduction.
- General description.
- Specific requirements.
- Appendices.
- Index.

PART-C

3a. i) Illustrate the different metrics used for specifying non-functional requirements.

Answer: Explanation 4M

Property	Measure
Speed	Processed transactions/second User/Event response time Screen refresh time
Size	M Bytes Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

ii)Design a template using structured natural language to capture the requirements of a system that is used for dispensing cash in a bank ATM.

Answer: Explanation 6M

Function :

Description :

Inputs :

Source :

Outputs:

Destination: **Main control loop**

Action

Requires:

Pre-condition:

Post-condition:

Side-effects:

3b i) Describe evolutionary development model and explain why programs that are developed during evolutionary development are likely to be difficult to maintain.

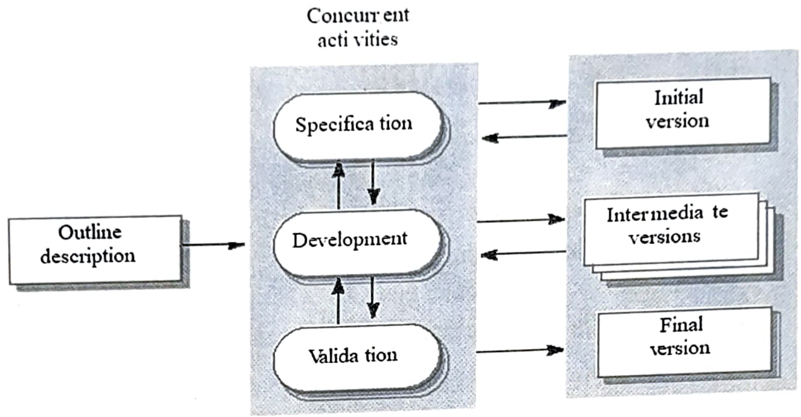
Answer: **Diagram-1M, Approaches -2M, problems-1M**

Exploratory development

- Objective is to work with customers and to evolve a final system from an initial outline specification. Should start with well-understood requirements and add new features as proposed by the customer.

Throw-away prototyping

- Objective is to understand the system requirements. Should start with poorly understood requirements to clarify what is really needed.

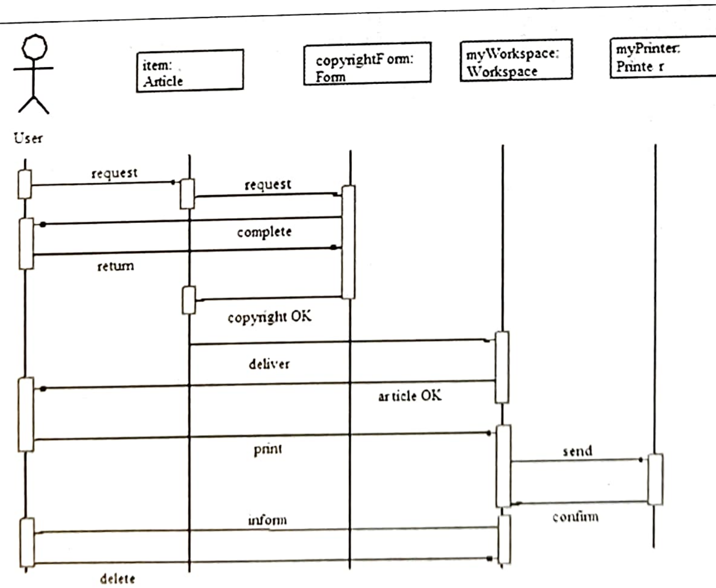


Problems

- Lack of process visibility;
- Systems are often poorly structured;
- Special skills (e.g. in languages for rapid prototyping) may be required.

ii) Design a sequence diagram for a system showing interaction for article printing.

Answer: Any relevant sequence diagram 6M



4a i) Write the scenario for carrying out online payment using credit card for a purchase of a book.

Answer : Scenario need to include - 4M

- A description of the starting situation;
- A description of the normal flow of events;
- A description of what can go wrong;
- Information about other concurrent activities;
- A description of the state when the scenario finishes.

ii) Giving reasons for your answers based on the type of system being developed, suggest the most appropriate generic software process model that might be used as a basis for managing the development of the following systems: 2*3=6M

a) An interactive system that allows flight passengers to find flight times from terminals installed in airports. - Interactive travel planning system with a complex user interface but which must be stable and reliable. An **incremental development** approach is the most appropriate as the system requirements will change as real user experience with the system is gained

b) The spelling checks and correcting function in a word processor. -The **Incremental Model** is suitable for a spelling check function as it allows gradual development and frequent updates based on user feedback.

c) A system to control anti-lock braking in a bike. – This is a safety-critical system so requires a lot of up-front analysis before implementation. It certainly needs a plan-driven approach to development with the requirements carefully analyzed. **A waterfall model** is therefore the most appropriate approach to use, perhaps with formal transformations between the different development stages.

4b) i) Suggest who might be the stakeholders in a university student record system. Justify, why it is almost inevitable that the requirements of different stakeholders will conflict in some way.

Stakeholders identification -2M, justification -2M

stakeholders in a university student record system: (or any other relevant stake holders)

1. **Students**: Primary users accessing and updating their academic and personal records.
2. **Faculty/Professors**: Users interacting with students' academic records to manage courses, grades, and performance.
3. **Academic Advisors**: Provide advice and track student academic progress.
4. **University Administration (Registrar's Office)**: Oversee all records, grades, course enrolments, and certifications.
5. **Admissions Office**: Manages student applications and records.

6. **IT Support/Systems Administrators**: Responsible for system maintenance, security, and functionality.
7. **Alumni**: Former students accessing historical data for various purposes.
8. **Financial Aid Office**: Manages and tracks financial aid and scholarships.
9. **Library Staff**: Track library usage and academic resources.
10. **Parents/Guardians**: Monitor and support the academic progress of students.

Justification:

Stakeholders don't know what they really want.
Stakeholders express requirements in their own terms.
Different stakeholders may have conflicting requirements.
Organisational and political factors may influence the system requirements.
The requirements change during the analysis process. New stakeholders may emerge and the business environment change

ii) Identify and Analyse the principal viewpoints which might be taken in to account and organize these using a viewpoint hierarchy diagram for an online food ordering and delivery system

Diagram -6M

Or any relevant diagram